Multimodal Emotion-Cause Pair Extraction with Holistic Interaction and Label Constraint

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The multimodal emotion-cause pair extraction (MECPE) task aims to detect the emotions, causes, and emotion-cause pairs from multimodal conversations. Existing methods for this task typically concatenate representations of each utterance from distinct modalities and then predict emotion-cause pairs directly. This approach struggles to effectively integrate multimodal features and capture the subtleties of emotion transitions, which are crucial for accurately identifying causes—thereby limiting overall performance. To address these challenges, we propose a novel model that captures holistic interaction and label constraint (HiLo) features for the MECPE task. HiLo facilitates cross-modality and cross-utterance feature interaction with various attention mechanisms, establishing a robust foundation for precise cause extraction. Notably, our model innovatively leverages emotion transition features as pivotal cues to enhance causal inference within conversations. The experimental results demonstrate the superior performance of HiLo, evidenced by an increase of more than 2% in the F1 score compared to existing benchmarks. Further analysis reveals that our approach adeptly utilizes multimodal and dialogue features, making a significant contribution to the field of emotion-cause analysis. Our code is publicly available at https://is.gd/MVdYmx.

CCS Concepts: • **Information systems** → **Multimedia information systems**.

Additional Key Words and Phrases: Multimodal Learning, Emotion Recognition

1 INTRODUCTION

Emotion-cause pair extraction (ECPE) [44, 58] aims to detect the emotions and the causes that lead to the emotions in text. This task provides insights for in-depth emotion analysis and empathetic response generation, and thus has been an active research area. Current approaches predominantly focus on analyzing text content, specifically targeting the detection of emotion and cause at the sentence level [20, 27]. However, this textual focus overlooks the rich context provided by multimodal interactions [15, 55] in daily life, where emotions are often expressed

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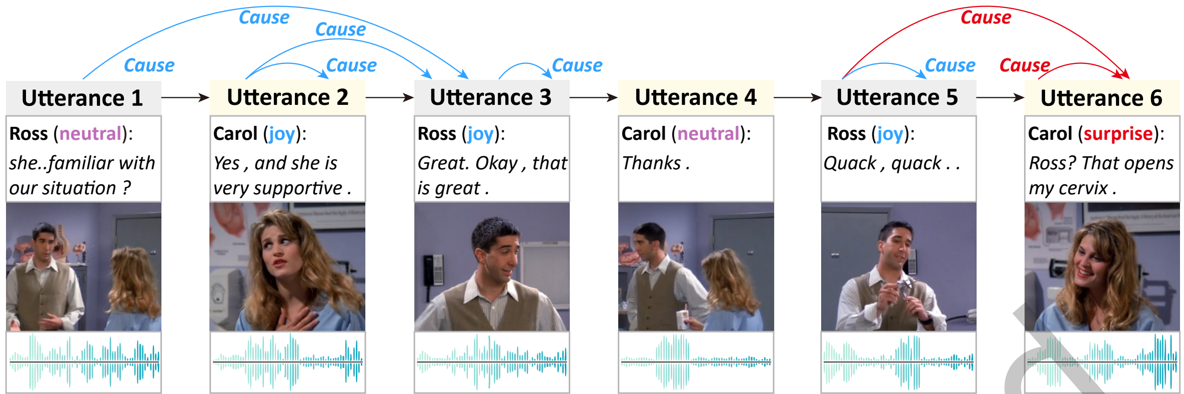


Fig. 1. Depiction of multimodal emotion-cause pair extraction in a dialogue scenario. Each utterance incorporates three distinct modalities—textual, visual, and auditory. Arrows represent the causative relationships, linking the eliciting utterances to the corresponding emotional responses.

through a combination of verbal and non-verbal cues [14, 57]. Solely text-based modalities may not capture the full spectrum of emotions and their causes, thereby limiting the performance of the ECPE task.

In light of this, the task of Multimodal Emotion-Cause Pair Extraction (MECPE) [52] was introduced to explore the potential of leveraging multimodal content for the improved extraction of emotion-cause pairs. As illustrated in Figure 1, given a conversation containing text, visual, and audio input, MECPE aims to detect all non-neutral emotion and cause utterances and extract utterance-level emotion-cause pairs (ECP). This task requires an understanding of the multimodal input and dialogue context [13, 28], then detecting emotions, causes, and emotion-cause pairs, thus presenting a significant challenge for accurate extraction.

To benchmark this task, a two-stage model was proposed in [52]. This method integrates the multimodal features and subsequently detects the emotions, causes, and emotion-cause pairs via an interaction matrix. Despite its efficacy, we identify two critical limitations that potentially hinder the performance. First, current models for multimodal emotion-cause pair extraction directly aggregate multimodal features to form utterance-level representations for the subsequent cause extraction stage. This process overlooks the differentiated influence that specific modalities might have on instigating certain emotions. For instance, in Figure 1, the *surprise* emotion in utterance #6 is primarily triggered by the action depicted in utterance #5 of Ross mimicking a frog, while the text and audio serve as auxiliary modalities. If these features are fused indiscriminately, the noise from less relevant modalities can hinder the extraction process. This realization underscores the need for feature interaction across both modality and utterance, a potentially promising approach for capturing more nuanced features and improving the precision of emotion-cause pair extraction.

Secondly, the existing methods conduct the dot product for each utterance pair to extract emotion-cause pairs. Such an approach fails to adequately exploit the transition of emotions in a conversation, a significant determinant for emotion-cause extraction. It is generally observed that a change in emotion between an individual's two successive utterances is often caused by the intervening utterances. For instance, in Figure 1, the transition from *neutral* in utterance #4 to *surprise* in utterance #6 for *Carol* appears to be triggered by utterance #5 from *Ross*. Here, #5 is an intervening utterance and plays a significant role in emotion-cause recognition. Statistical data from the annotated dataset lend further support to this observation: when emotions differ between successive utterances from the same speaker, the intervening utterance is recognized as the cause in 55% of cases. This number represents a notable increase compared to the 33% when emotions between successive utterances from the same speaker remain consistent. Such findings highlight the importance of monitoring emotional transitions as pivotal reference points in emotion-cause extraction. In addition, as the distance between utterances expands, the probability of them forming an emotion-cause relation correspondingly decreases. This underlines the critical role played by the relative position of utterances within this task.

To address the existing issues, we propose a method based on holistic interaction and emotional label constraint (HiLo) modeling for MECPE. Our approach begins by extracting multimodal features for each utterance using three specialized feature extractors. Subsequently, we conduct the cross-utterance and cross-modality feature interaction with a transformer-like self-attention mechanism from three distinct perspectives: global, speaker, and replying. Following the acquisition of the representation from the attention layer, we carry out emotion classification, cause classification, and ECPE. Importantly, we incorporate the discrepancy between successive utterances from the same speaker into the ECPE process, ensuring that greater discrepancies increase the likelihood of the middle utterance being a cause for the latter utterance. Additionally, we equip utterance representations in the ECPE process with Rotary Position Embeddings (RoPE) [48] to better model the influence of relative distance. In essence, our model enhances fine-grained interaction and leverages the inherent dynamics of emotion transition in conversations, thereby enabling comprehensive modeling of multimodal and dialogue features for the ECPE task.

To validate the performance of our model, we conduct experiments on the MECPE dataset and assess the scores for emotion, cause, and ECPE tasks. The results show that our model outperforms existing approaches across all three tasks, achieving a notable improvement of more than 2% in the F1 score for the ECPE task. Further analysis illustrates that our model effectively leverages both multimodal and dialogue features for the MECPE task. Overall, our key contributions can be summarized as follows:

- We introduce the HiLo model tailored for multimodal emotion-cause pair extraction, leveraging cross-utterance and cross-modality interactions to enhance utterance representation.
- We identify the pivotal role of emotion transition in conversational ECPE and, accordingly, design a novel label constraint mechanism to capture these transitions.
- Experimental results verify the superiority of HiLo, and the in-depth analysis emphasizes the efficacy of our fine-grained feature interaction and label constraint approach.

2 RELATED WORK

2.1 Emotion-Cause Pair Extraction

Emotion-Cause Pair Extraction (ECPE) involves the identification of emotion-cause pairs within documents [58] or dialogues [44], where both the emotion and the cause find expression within clauses. The original ECPE model, mainly oriented towards documents, followed a two-step framework: first, extracting emotions and causes separately; then, combining the identified pairs [58]. Subsequent studies have been proposed to enhance the task's performance from various perspectives, such as enhancing sentence interactions [10, 53, 64], capturing contextual features [1, 6, 12, 35], leveraging boundary features [7, 18, 32], integrating external knowledge [59, 67], and aligning multi-task features [4, 5, 54, 70, 71], among others. Another stream of research focuses on conversational emotion-cause pair extraction. The benchmark for ECPE in conversation primarily comprises RECCON [44], HECE-DD [42], and ECPEC [31]. Several models have explored graph neural networks [19] and commonsense knowledge [68] to infer the causes of emotions in conversations. Notwithstanding their commendable performance, these models predominantly concentrate on unimodal data, limiting their effectiveness in real-world scenarios that frequently encompass multimodal data. Hence, this paper focuses on multimodal conversational emotion-cause pair extraction.

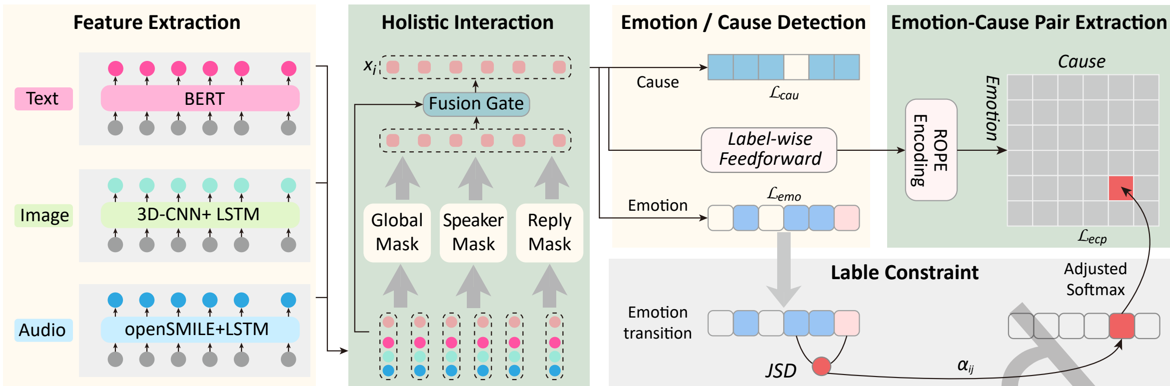


Fig. 2. The overall framework of our proposed model. First, multimodal features are extracted via three distinct feature extractors. Following this, we facilitate cross-utterance and cross-modality feature interactions using a multi-head attention mechanism. The next step involves classifying the emotion/cause category for each utterance. The final stage is dedicated to extracting emotion-cause pairs, guided by an emotion-transition constraint. JSD is the Jensen–Shannon divergence.

2.2 Multimodal Emotion Recognition in Conversation

The task of multimodal emotion recognition in conversation (MMERC) focuses on the analysis of emotion categories within conversations. Oriented towards this task, numerous studies have explored various multimodal fusion techniques, such as attention-based fusion [22, 37, 46, 47, 62, 63], graph-based fusion [24, 30, 39, 45, 69], hierarchical level fusion [3, 8], and enhanced interaction fusion [26, 29, 60, 61, 65]. Other methods explore the influence of label schemas from different perspectives to tackle the task, such as multimodal learning [2, 25, 66], contrastive learning [23, 50], and specialized loss functions [33]. Nevertheless, despite their efficacy in understanding emotion categories, they cannot be directly applied to the emotion-cause pair extraction task without adaptations. Thus, while building upon the foundational work of emotion recognition models, this paper also uniquely concentrates on enhancing the interaction between utterances to elucidate their causative relationships.

3 TASK DEFINITION

Before delving into the specifics of our methodologies, it is imperative to define the task at hand clearly. The task of MECPE focuses on analyzing a conversation $D = \{(s_1, u_1), (s_2, u_2), \ldots, (s_n, u_n)\}$, where $u_i = (t_i, a_i, v_i)$ denotes the i-th utterance, incorporating textual, auditory, and visual modalities, and s_i denotes the corresponding speaker. This task seeks to identify the **emotion label** and the **cause label** for each utterance, in addition to extracting all **emotion-cause pairs** within the given conversation. The emotion label for u_i corresponds to categories such as *joy*, *sadness*, etc. The cause related to u_i is identified by a binary label, signifying whether u_i incites an emotion in another utterance. The pair (u_i, u_j) indicates that u_j is the causative factor for the emotion in u_i.

4 METHOD

This section starts with feature extraction. Then, we outline the framework in order: holistic interaction, emotion/cause detection, emotion-cause pair extraction, and finally, label constraint. The concluding segment of this section details the loss function used in the learning stage.

4.1 Feature Extraction

Given D, we first employ three distinct feature extractors to obtain the raw features for each utterance. For the textual modality, considering the powerful semantic representation capability of Pretrained Language Models, we utilize BERT [9] in this paper to encode text content. We initially concatenate all utterances into a long sequence, separated by CLS and SEP tokens, and then feed this sequence into BERT to obtain a deep contextualized representation $\boldsymbol{t}_i^b$ for each u_i:

$$\text{Input} = \{[CLS], w_{00}, \cdots, [CLS], w_{10}, \cdots, [SEP]\}\ , \tag{1}$$

$$\boldsymbol{h}_{cls,0}, \cdots, \boldsymbol{h}_{cls,1}, \cdots, \boldsymbol{h}_{sep} = \textbf{BERT}(\text{Input})\ , \tag{2}$$

$$\boldsymbol{t}_i^b = \boldsymbol{h}_{cls,i}\ , \tag{3}$$

where w_{ij} denotes the j-th token in the i-th utterance, and $\boldsymbol{h}_{ij}$ is the representation of the corresponding token. We represent $\boldsymbol{t}_i^b$ using the vector of the i-th CLS token.

For the audio modality, we initially employ the OpenSMILE toolkit [11] to extract the raw features $\boldsymbol{a}_i^r$ for a_i. Subsequently, a Bidirectional Long Short-Term Memory (Bi-LSTM) [21, 38] is utilized to acquire the contextualized feature as follows:

$$\boldsymbol{a}_1^b, \boldsymbol{a}_2^b, \cdots, \boldsymbol{a}_n^b = \textbf{BiLSTM}_a(\boldsymbol{a}_1^r, \boldsymbol{a}_2^r, \cdots, \boldsymbol{a}_n^r)\ , \tag{4}$$

where $\boldsymbol{a}_i^b$ is the updated audio representation of the i-th utterance.

Similarly, for the visual modality, we leverage a 3D-CNN [49] to extract the raw feature $\boldsymbol{v}_i^r$, subsequently obtaining the contextualized representation $\boldsymbol{v}_i^b$ for u_i:

$$\boldsymbol{v}_1^b, \boldsymbol{v}_2^b, \cdots, \boldsymbol{v}_n^b = \textbf{BiLSTM}_v(\boldsymbol{v}_1^r, \boldsymbol{v}_2^r, \cdots, \boldsymbol{v}_n^r)\ . \tag{5}$$

Thereafter, we fuse the representations of three modalities with speaker embedding to obtain a representation at the utterance level:

$$\boldsymbol{x}_i^b = \boldsymbol{t}_i^b + \boldsymbol{a}_i^b + \boldsymbol{v}_i^b + \boldsymbol{e}_i\ , \tag{6}$$

where $\boldsymbol{x}_i^b$ denotes the representation of u_i, $\boldsymbol{e}_i$ represents the randomly initialized speaker embedding for u_i. In detail, we assign an index to each speaker and construct a mapping matrix, with each index i corresponding to a vector $\boldsymbol{e}_i$ in the matrix. Then, during the training process, the weight of the matrix can be updated, and thus the parameters in the embedding vector reflect the features of the speaker role.

4.2 Holistic Interaction

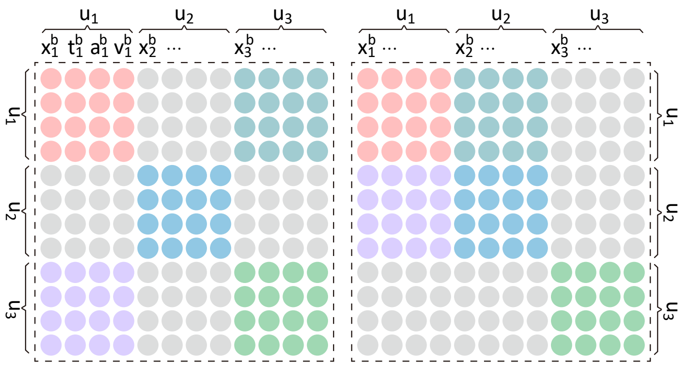
To capture the fine-grained interaction between utterances, we devise a holistic (cross-utterance and cross-modality) interaction mechanism based on the Transformer block [51].

Initially, we concatenate the representations of all utterances for each modality, along with the utterance representations themselves, to form an elongated sequence.

$$\boldsymbol{X} = [\boldsymbol{x}_1^b, \boldsymbol{t}_1^b, \boldsymbol{a}_1^b, \boldsymbol{v}_1^b, \boldsymbol{x}_2^b, \cdots, \boldsymbol{v}_n^b]\ . \tag{7}$$

Then we build the mask matrix $\boldsymbol{M}^m \in \mathbb{R}^{4n\times 4n}$ to denote the interaction between different features in $\boldsymbol{X}$, where $m \in \{g, s, r\}$ represents interaction from the *global*, *speaker*, and *reply* perspective, respectively. Here is how M^m is determined:

- **Global**: For every $1 \leq i, j \leq 4 \cdot n$, we assign $\boldsymbol{M}_{ij}^g = 1$.
- **Speaker**: $\boldsymbol{M}_{ij}^s = 1$ is established when $s_{\underline{i}} = s_{\underline{j}}$, signifying both positions are associated with the same speaker.
- **Reply**: If $u_{\underline{i}}$ acts as a reply to $u_{\underline{j}}$ or the converse is true, we assign $\boldsymbol{M}_{ij}^r = 1$.



(a) Reply Mask (b) Speaker Mask

Fig. 3. Illustration of the reply and speak masks: u_3 replies to u_1, and u_1 and u_2 are from the same speaker. Gray circles symbolize the lack of interaction between the feature pair, whereas other colors represent a valid interaction. Note that the global mask is not illustrated since interactions are inherent between every pair of features.

Here, $\underline{i}$ and $\underline{j}$ are defined as $\underline{i} = \lfloor i/4 \rfloor$ and $\underline{j} = \lfloor j/4 \rfloor$, respectively, which represent the actual utterance index. Then based on the $\boldsymbol{M}^m$, we conduct a multi-head self-attention mechanism over the sequence $\boldsymbol{X}$ to obtain the comprehensive interaction result:

$$\mathrm{H}_i^m = \mathrm{Softmax}\left(\frac{(\boldsymbol{X}\boldsymbol{W}_q)(\boldsymbol{X}\boldsymbol{W}_k)^T \cdot \boldsymbol{M}^m}{\sqrt{\mathrm{d}}}\right)(\boldsymbol{X}\boldsymbol{W}_v)\,, \tag{8}$$

$$\mathbf{MHA}(\boldsymbol{X}, m) = [\mathrm{H}_1^m; \cdots; \mathrm{H}_h^m]\boldsymbol{W}_O\,, \tag{9}$$

where d denotes the dimension of $\boldsymbol{X}$, and $\boldsymbol{M}^m \in \mathbb{R}^{4n \times 4n}$ acts as a mask matrix managing the interaction between elements at corresponding positions. $\boldsymbol{W}_Q$, $\boldsymbol{W}_K$, $\boldsymbol{W}_V$, and $\boldsymbol{W}_O$ are learnable parameters. $\boldsymbol{H}_i^m$ denotes the result of i-th head. In combination with Figure 3, when $\boldsymbol{M}_{ij}^m = 1$, there exists an interaction between the i-th and j-th feature. Conversely, $\boldsymbol{M}_{ij}^m = 0$ signifies an absence of interaction.

Then, we slice the sequence to obtain the overall representation for each utterance:

$$\boldsymbol{x}_1^m, \cdots, \boldsymbol{x}_n^m = \mathbf{MHA}(\boldsymbol{X}, m)[1 : 4n+1 : 4]\,. \tag{10}$$

The notation $(\cdot)$[1:4n+1:4] signifies a slicing operation that extracts elements from the list, starting from the first index and extending to the n-th index, picking every fourth element in this range. Thus, $\boldsymbol{x}_i^m$ serves as the updated representation for u_i.

Then, we utilize a max-pooling layer to fuse the representations from the three perspectives:

$$\boldsymbol{x}_i^o = \mathbf{MaxPooling}([\boldsymbol{x}_i^g, \boldsymbol{x}_i^s, \boldsymbol{x}_i^r])\,, \tag{11}$$

where $\boldsymbol{x}_i^o \in \mathbb{R}^d$ is the updated representation for u_i. To maintain the original feature, we use a gating mechanism to adaptively fuse the original feature with the updated one:

$$\mathcal{F} = \sigma(\boldsymbol{W}_f[\boldsymbol{x}_i^b, \boldsymbol{x}_i^o] + \boldsymbol{b}_f)\,, \tag{12}$$

$$\boldsymbol{x}_i = \mathcal{F} * \boldsymbol{x}_i^b + (1 - \mathcal{F}) * \boldsymbol{x}_i^o\,, \tag{13}$$

where σ represents the Sigmoid activation function. $\boldsymbol{W}_f \in \mathbb{R}^{d\times 2d}$ and $\boldsymbol{b}_f \in \mathbb{R}^d$ are the parameters, and $\boldsymbol{x}_i$ is the representation for $\boldsymbol{u}_i$.

4.3 Emotion/Cause Detection

After obtaining the learned representation, we proceed with emotion detection, which employs a feed-forward neural network and softmax function:

$$\boldsymbol{h}_i^e = \mathrm{FFNN}_e(\boldsymbol{x}_i)\ , \tag{14}$$

$$\boldsymbol{p}_i^e = \mathrm{Softmax}(\boldsymbol{h}_i^e)\ , \tag{15}$$

where FFNN_e is a two-layer feed-forward network, and $\boldsymbol{p}_i^e$ is the predicted probability for the emotion label. Similarly, we conduct cause detection as follows:

$$\boldsymbol{p}_i^c = \mathrm{Softmax}(\mathrm{FFNN}_c(\boldsymbol{x}_i))\ , \tag{16}$$

where $\boldsymbol{p}_i^c$ is the predicted probability for the cause label.

4.4 Emotion-Cause Pair Extraction

For emotion-cause pair detection, we start by obtaining two label-wise representations for each utterance:

$$\boldsymbol{x}_i^p = \mathrm{FFNN}_{pos}(\boldsymbol{x}_i)\ , \tag{17}$$

$$\boldsymbol{x}_i^n = \mathrm{FFNN}_{neg}(\boldsymbol{x}_i)\ , \tag{18}$$

where $\boldsymbol{x}_i^p$ and $\boldsymbol{x}_i^n$ are the representations for positive and negative labels, respectively.

Next, we incorporate relative distance encoding into these representations using RoPE [48]:

$$\hat{\boldsymbol{x}}_i^p = \boldsymbol{R}(\theta, i)\boldsymbol{x}_i^p\ , \tag{19}$$

$$\hat{\boldsymbol{x}}_i^n = \boldsymbol{R}(\theta, i)\boldsymbol{x}_i^n\ , \tag{20}$$

where $\boldsymbol{R}(\theta, i)$ is the relative position matrix parameterized by θ and position i, and both $\hat{\boldsymbol{x}}_i^p$ and $\hat{\boldsymbol{x}}_i^n$ share the same shape as $\boldsymbol{x}_i$.

For each pair of utterances, we can then compute the score for them being or not being an emotion-cause pair:

$$p_{ij}^p = (\hat{\boldsymbol{x}}_i^p)^\top \hat{\boldsymbol{x}}_j^p\ ,\ p_{ij}^n = (\hat{\boldsymbol{x}}_i^n)^\top \hat{\boldsymbol{x}}_j^n\ , \tag{21}$$

where p_{ij}^p is a scalar denoting the score that u_i and u_j form an emotion-cause pair, whereas p_{ij}^n denotes the score of them not forming a pair, with $(\cdot)^\top$ representing the transpose operation.

Finally, we can compute the binary classification probability:

$$\boldsymbol{p}_{ij}^{ec} = \mathrm{Softmax}_c([p_{ij}^n, p_{ij}^p])\ , \tag{22}$$

where $\boldsymbol{p}_{ij}^{ec}$ denotes the probability that u_i and u_j constitute an emotion-cause pair, with u_j being the cause of u_i. $\mathrm{Softmax}_c$ is a softmax function adjusted by a label constraint strategy, which will be discussed in the following section.

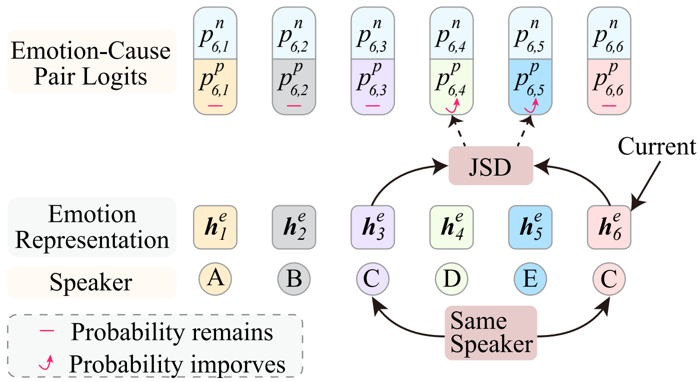


Fig. 4. Illustration of the label constraint mechanism influenced by emotion transition. We consider u_6 as the current utterance and compute the probability that a preceding utterance is the cause of u_6. Notably, u_3 shares the same speaker as u_6, hence the label constraint mechanism, determined by the emotion discrepancy between $\boldsymbol{h}^e_3$ and $\boldsymbol{h}^e_6$, is applied to the pair probabilities of (u_6, u_5) and (u_6, u_4). JSD is the Jensen–Shannon divergence.

4.5 Label Constraint

In addition to basic extraction, we further consider the transition of the same speaker's emotion as a soft constraint and incorporate it into the softmax function to enhance emotion-cause pair extraction. For the i-th utterance u_i in D, we define $F(i)$ as the index of the most recent preceding utterance by the same speaker as u_i:

$$F(i) = \begin{cases} \max\{j \in [0, i-1] : s_j = s_i\}, & \text{if } j \text{ exists}, \\ i, & \text{otherwise}. \end{cases} \tag{23}$$

For instance, in Figure 4, $F(6) = 3$ and $F(5) = 5$, among others.

Then, for each utterance pair, we can compute a factor score:

$$\alpha_{ij} = \begin{cases} \beta \cdot \mathrm{JSD}(\boldsymbol{h}^e_i \,||\, \boldsymbol{h}^e_{F(i)}), & F(i) \le j < i\,, \\ 0, & otherwise\,. \end{cases} \tag{24}$$

Here, β is a positive-valued hyperparameter. $\boldsymbol{h}^e_i$ is the emotional representation in Eq. (14) for u_i. JSD is the Jensen–Shannon divergence [34], which evaluates the discrepancy between two vectors and ranges from 0 to $\ln(2)$. A higher discrepancy between $\boldsymbol{h}^e_i$ and $\boldsymbol{h}^e_{F(i)}$ indicates a larger JSD value, and thus a large α_{ij}. We then incorporate α_{ij} as a parameter into the adjusted softmax:

$$\mathrm{Softmax}_c([p^n_{ij}, p^p_{ij}]) = \mathrm{Softmax}(p^n_{ij}\,, \exp(\alpha_{ij}) p^p_{ij})\,. \tag{25}$$

When α_{ij} is larger, it indicates a higher probability that u_i and u_j form an emotion-cause pair. Conversely, if $\alpha_{ij} = 0$, the model defaults to using the regular softmax function. With this modification, an utterance between two utterances from the same speaker is more likely to be the cause of the latter utterance, particularly if the emotion transition of the individual is significant. For utterances outside this scope, where $j < F(i)$ or $j > i$ causing $\alpha_{ij} = 0$, predictions rely on the traditional softmax and remain uninfluenced.

Table 1. Dataset statistics for train, dev, test set, and all set, respectively.

Item	Train	Dev	Test	All
Conversations	1,001	112	261	1,374
Utterances	9,966	1,087	2,566	13,619
Emotions	5,577	668	1,445	7,690
Emotion-Cause Pairs	7,055	866	1,873	9,794
Emotion (with Cause)	5,123	614	1,344	7,081
Emotion (1 Cause)	3,566	421	927	4,914
Emotion (2 Cause)	1,252	145	327	1,724
Emotion (3 Cause)	244	38	72	354

4.6 Learning

To facilitate learning, we define losses for each task.

$$\mathcal{L}_{emo} = -\frac{1}{n}\sum_{i}^{n} \log \boldsymbol{p}_i^e(y_i^e) \,, \tag{26}$$

$$\mathcal{L}_{cau} = -\frac{1}{n}\sum_{i}^{n} \log \boldsymbol{p}_i^c(y_i^c) \,, \tag{27}$$

$$\mathcal{L}_{ecp} = -\frac{2}{n(n+1)}\sum_{i=1}^{n}\sum_{j=1}^{i} \log \boldsymbol{p}_{ij}^{ec}(y_{ij}^p) \,. \tag{28}$$

The final loss function, combining the three tasks, is:

$$\mathcal{L} = \mathcal{L}_{emo} + \mathcal{L}_{cau} + \mathcal{L}_{ecp} \,. \tag{29}$$

During training, we minimize $\mathcal{L}$ to optimize the model.

5 EXPERIMENT

5.1 Dataset

We utilize the MECPE dataset [44] for our experiments. Originating from the television series *Friends*, this dataset encompasses multi-turn conversation scripts, along with audio and video excerpts. The data contains annotations for emotion categories, binary cause indicators for each utterance, and all emotion-cause utterance pairs in the dialogues. As is an expanded version of the MELD dataset [43], MECPE contains 1,374 conversations, and 13,619 utterances. The detailed statistics for the dataset can be found in Table 1.

5.2 Evaluation Metrics

Consistent with previous methodologies [52], we evaluate our model's performance in emotion-cause pair extraction using *precision* (P) , *recall* (R), and *F1 score* (F1). These metrics are specifically defined for each evaluated task. For emotion-cause pair extraction:

$$\mathrm{P} = \frac{\sum \text{correct pairs}}{\sum \text{predicted pairs}}, \tag{30}$$

$$\mathrm{R} = \frac{\sum \text{correct pairs}}{\sum \text{golden pairs}}, \tag{31}$$

$$\mathrm{F1} = \frac{2 \times \mathrm{P} \times \mathrm{R}}{\mathrm{P} + \mathrm{R}}, \tag{32}$$

where "correct pairs" refer to accurately identified emotion-cause pairs, "predicted pairs" denote the model's predictions, and "golden pairs" are the annotated instances in the dataset.

For emotion detection, we apply these metrics to assess the accuracy of identified non-neutral emotions:

$$\mathrm{P} = \frac{\sum \text{correct non-neutral emotions}}{\sum \text{predicted non-neutral emotions}}, \tag{33}$$

$$\mathrm{R} = \frac{\sum \text{correct non-neutral emotions}}{\sum \text{golden non-neutral emotions}}, \tag{34}$$

$$\mathrm{F1} = \frac{2 \times \mathrm{P} \times \mathrm{R}}{\mathrm{P} + \mathrm{R}}. \tag{35}$$

Similarly, for cause detection:

$$\mathrm{P} = \frac{\sum \text{correct causes}}{\sum \text{predicted causes}}, \tag{36}$$

$$\mathrm{R} = \frac{\sum \text{correct causes}}{\sum \text{golden causes}}, \tag{37}$$

$$\mathrm{F1} = \frac{2 \times \mathrm{P} \times \mathrm{R}}{\mathrm{P} + \mathrm{R}}. \tag{38}$$

While our primary metrics focus on overall performance, they do not differentiate between individual emotion labels. To provide a more nuanced assessment, we also implement category-level *precision*, *recall*, and *F1 score* metrics for each specific emotion. We computed the weighted average scores Avg.6 and Avg.4 as the overall assessment metrics. The former is the weighted average for all emotions, and the latter excludes two categories with fewer instances, namely, 'disgust' and fear'.

5.3 Baselines

For comparison, our baselines include the heuristic and deep learning models proposed in the original MECPE paper [52]. Since large language model [16, 17, 41, 56] has achieve leading performance in vairous information detection task, we also compare the performance of LLMs on the MECPE task.

Heuristic Approach Heuristic methods exploit inherent patterns related to the task to extract emotion-cause pairs.

Deep Learning Approach This approach employs a two-step framework: initially detecting emotions and causes, then extracting emotion-cause pairs.

LLM-based Zero Shot We utilize LLMs, including GPT-3.5 [41] and GPT-4 [40], for zero-shot emotion-cause pair extraction. The prompt format is detailed in Figure 5.

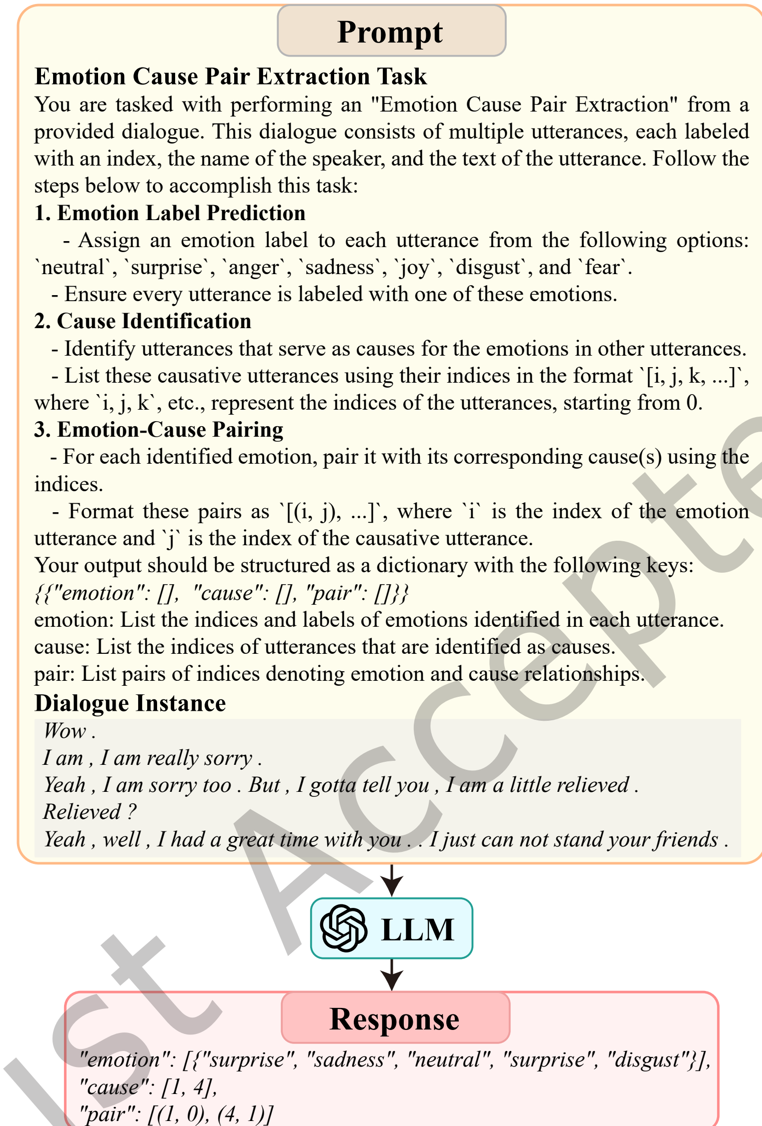


Fig. 5. Prompt designed to guide ChatGPT in the task of extracting emotion-cause pairs from dialogue, illustrating the specific instructions, input, and output formats used.

5.4 Settings

In our model, we utilize the 'bert-base-cased' version of BERT [9] as our pretrained language model, which features 12 layers and provides vectors of 768 dimensions. For audio data encoding, we opt for OpenSMILE [11], while 3D-CNN [49] is used for video data encoding. During the training process, we implement different learning rates for BERT and non-BERT layers, set at 1e-3 and 1e-5, respectively. We employ AdamW [36] with ϵ set to 1e-7 as the optimizer, in conjunction with a linear learning rate scheduler. The training procedure includes a

Table 2. Experiment results on the MECPE task. We present Precision/Recall/F1 scores for emotion detection, cause detection, and ECPE. We do not distinguish the emotion category in this table, and scores with wavelines denote the second-best baselines.

Methods		Emotion Detection			Cause Detection			Pair Extraction		
		P	R	F1	P	R	F1	P	R	F1
Heuristic Approach	$E_{Pred.}$ + $C_{Bern.}$	73.62	79.68	76.48	54.91	50.28	52.44	36.99	26.77	31.01
	$E_{Anno.}$ + $C_{Bern.}$	100.00	100.00	100.00	64.20	54.85	59.15	49.58	33.07	39.67
	$E_{Pred.}$ + $C_{Multi.}$	73.62	79.68	76.48	54.88	50.22	52.39	36.94	26.71	30.96
	$E_{Anno.}$ + $C_{Multi.}$	100.00	100.00	100.00	64.17	54.83	59.13	49.51	33.03	39.63
Deep Learning	MECPE-2steps	77.17	81.36	79.10	67.47	73.19	70.13	57.64	48.72	52.71
	+ Audio	76.91	81.68	79.17	67.25	73.91	70.27	57.13	50.34	53.48
	+ Video	77.10	81.18	79.03	68.10	72.51	70.18	**57.77**	49.53	53.21
	+ Audio + Video	77.45	81.10	79.16	**68.42**	72.43	70.27	57.47	49.81	53.20
LLMs	GPT-3.5	57.75	65.08	61.19	39.87	29.30	33.78	8.95	51.23	15.24
	GPT-4	69.28	75.44	72.23	51.44	36.74	42.86	12.56	52.34	20.26
Our Model	HiLo (Text)	74.53	81.80	77.99	65.26	80.23	71.97	51.91	54.33	53.09
	+ Audio	**78.31**	77.71	78.01	66.32	79.17	72.18	51.50	56.65	53.95
	+ Video	76.03	80.14	78.03	61.98	**85.89**	72.01	54.60	52.71	53.64
	+ Audio + Video	75.46	**83.66**	**79.35**	65.72	80.31	**72.28**	51.78	**59.69**	**55.45**

Table 3. Ablation study conducted on holistic interaction, label constraint strategy, and RoPE encoding. We provide the precision, recall, and F1 scores for emotion-cause pairs, as well as the weighted average score across different emotion categories. Note that Avg.4 is calculated excluding *Disgust* and *Fear* due to their comparatively small sample sizes.

Methods	P	R	F	Avg.6	Avg.4
HiLo	51.78	59.69	**55.45**	33.04	**35.81**
-Speaker	49.47	60.01	54.23	32.28	35.06
-Reply	49.23	58.40	53.43	32.40	35.06
-Global	53.12	53.97	53.54	31.82	33.96
-HI	**57.97**	48.69	52.93	30.48	32.79
-LO	56.17	53.23	54.66	**32.79**	34.82
-RoPE	48.74	**60.06**	53.81	31.39	33.83

batch size of 8, a maximum of 20 epochs, and an early stopping mechanism triggered by a lack of improvement over five consecutive epochs on the development set. We conduct all of our experiments on an Ubuntu server equipped with an RTX 3090 GPU. It takes around 30 minutes to complete both the training and inference phases on the entire dataset.

5.5 Main Result

Table 2 presents our experimental results. For the well-developed task of emotion detection, our HiLo model achieves a slight improvement over the baseline, with an increase of 0.18 percent. Remarkably, HiLo registers a substantial 2.01 percent improvement in the F1 score for cause detection, which partly contributes to a notable 1.97 percent boost in the F1 score for emotion-cause pair extraction. Given that cause detection and more importantly emotion-cause pair extraction, constitute the essence of the MECPE task, this significant improvement underscores the superiority of our model compared to the baseline approaches.

Table 4. The experimental results of MECPE. We present the F1 scores for pair extraction across each emotion category and their weighted F1 scores. Note that we exclude *Disgust* and *Fear* from Avg.4 as their numbers are relatively small. The abbreviations 'Pred.', 'Bern.', 'Anno.', and 'Multi.' denote 'Prediction', 'Bernoulli', 'Annotation', and 'Multinomial', respectively. The terms 'Disgust', 'Sadness', and 'Surprise' are abbreviated as 'Disg.', 'Sad.', and 'Surp.', respectively.

Methods		Pair Extraction						Weighted	
		Anger	Disg.	Fear	Joy	Sad.	Surp.	Avg.6	Avg.4
Heuristic Approach	$E_{Pred.}$ + $C_{Bern.}$	16.69	2.95	2.27	22.53	9.94	22.85	17.32	18.63
	$E_{Anno.}$ + $C_{Bern.}$	17.55	3.04	2.41	24.63	10.80	24.51	18.64	20.06
	$E_{Pred.}$ + $C_{Multi.}$	16.62	2.90	2.28	22.50	9.96	22.89	17.31	18.61
	$E_{Anno.}$ + $C_{Multi.}$	17.59	3.03	2.41	24.57	10.74	24.53	18.63	20.04
Deep Learning	MECPE-2steps	24.39	0.00	0.71	38.84	21.60	40.24	29.32	31.92
	Deep + Audio	26.30	0.46	0.50	39.62	22.73	41.35	30.47	33.15
	+ Video	22.68	0.19	0.36	38.55	20.58	39.99	28.57	31.11
	+ Audio + Video	25.08	0.09	1.01	38.49	22.36	40.64	29.62	32.24
Our Model	HiLo (Text)	28.95	8.82	**14.89**	39.75	19.87	**44.15**	32.14	34.05
	+ Audio	28.85	1.77	3.51	41.38	23.50	43.67	32.39	35.09
	+ Video	**30.02**	**13.07**	5.13	41.60	**26.92**	41.78	**33.61**	35.68
	+ Audio + Video	29.17	1.68	3.12	**42.58**	25.10	43.55	33.04	**35.81**

Moreover, the performance of our model significantly differs across various multimodal inputs, particularly for the emotion-cause pair extraction task. The model employing solely text modality performs the worst, lagging behind combinations of text+audio or text+video, while the fusion of all three modalities delivered the best results. Interestingly, unlike the baseline model, the performance of HiLo escalates as more modalities are included, showcasing its ability to harness the power of fine-grained multimodal interactions effectively. This, in turn, significantly optimizes the interaction and integration of multimodal inputs, contributing to the stable performance improvement in the MECPE task.

In addition, we find that LLMs show average performance in this task. Despite LLMs' comparable performance in emotion detection, they fall short in cause detection and ECPE compared to our model. This indicates that there is still a long way to go for LLMs to understand the complex causes of emotions in dialogue scenarios.

5.6 Ablation Study

To evaluate the contributions of each module in our model, we performed an ablation study. As shown in Table 3, we observed that the holistic interaction is essential for our task. The removal of the holistic interaction module led to a decline in the F1 score of more than 2.5 percent. Moreover, eliminating each sub-interaction also impacted the overall performance to various degrees. Secondly, the removal of the label constraint and RoPE module resulted in substantial performance drops of 0.79 and 1.64 percent, respectively. In summary, the ablation study affirmed the effectiveness of each module in our model and the rationality of its design. The detailed mechanisms of how each module influences the extraction will be further explored in the subsequent section.

5.7 Category-level Analysis

To gain a deeper understanding of our model's performance, we examined the extraction of emotion-cause pairs across different emotion categories. As shown in Table 4, we observed a weaker performance in the extraction of pairs associated with the emotions of disgust and fear. This could likely be attributed to the relatively small sample size for these categories, which hindered the model's ability to learn effective features. In stark contrast, our model excelled for the other four emotion categories, elevating the overall metric by more than two percent

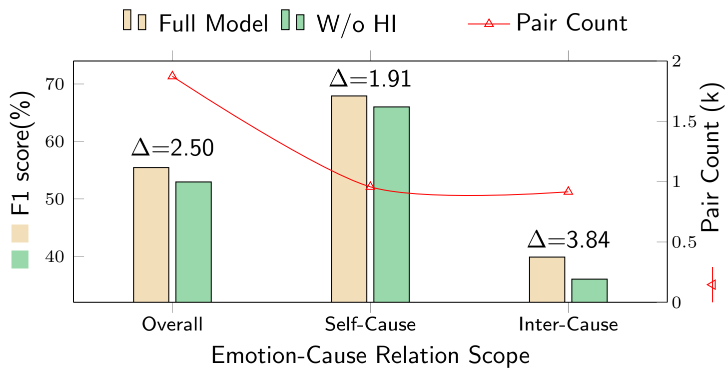


Fig. 6. Results of emotion-cause pair extraction, categorized into all pairs, intra-utterance pairs (self-cause, where the cause is the utterance itself), and inter-utterance pairs (inter-cause, where the cause is a different utterance). The symbol Δ denotes the gap in each scenario between the full model and the model without the holistic interaction module.

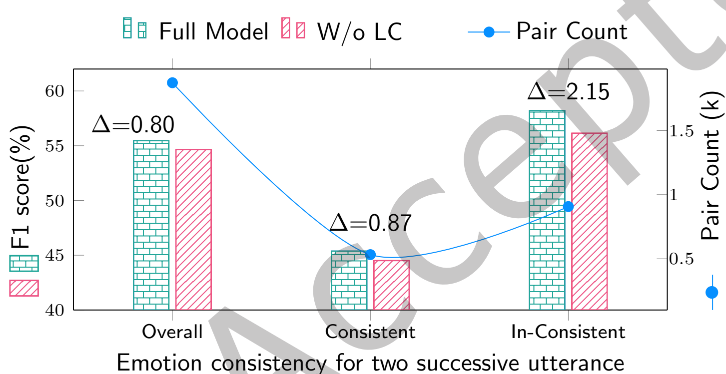


Fig. 7. Experimental results grouped by all pairs, consistent and inconsistent pairs. Consistent pairs are those where an utterance and the nearest preceding utterance from the same speaker share the same emotion. Conversely, inconsistent pairs refer to utterances where emotions differ. We also provide the pair count for each group. The sum of the last two groups does not equal the overall result, as some utterances lack a preceding utterance from the same speaker, which are not included in the count. The symbol Δ denotes the gap in each scenario between the full model and the model without the label constraint module.

relative to the baseline models. With improvements of 3.14 in Avg.6 and 2.66 in Avg.4, our model affirms its efficacy and stability. Finally, different from baseline models, our model reached its peak performance with text + audio + video input. The incorporation of the audio and visual modality did indeed amplify the performance, highlighting the prowess of our model in seamlessly assimilating multimodal inputs, especially for the intricate task of emotion-cause extraction across diverse emotion categories.

5.8 In-depth Analysis

To better understand the working mechanism of HiLo, we conducted further analysis to answer several key questions.

Q1: How does holistic interaction influence the extraction result?

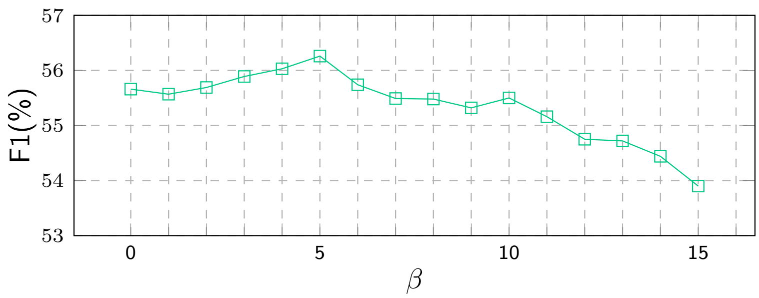


Fig. 8. Variation of the F1 score for Emotion-Cause Pair Extraction (ECPE) on the development set with the change of the β in Eq. (24).

A1: We assessed the impact of holistic interaction on emotion-cause pair extraction and compared the performance of emotion-cause pairs intra and inter-utterances. Figure 6 demonstrates that holistic interaction enhances emotion-cause pair extraction performance, especially for inter-utterance pairs. Our model increases the overall score by 2.5 percent, with a notable improvement of 3.84 percent in inter-utterance pairs—double the 1.9 percent enhancement observed in intra-utterance pairs. These results substantiate that holistic interactions facilitate effective relationships between distinct utterances, thereby bolstering the performance of emotion-cause pair extraction.

Q2: Does the label constraint mechanism effectively leverage emotion transitions for emotion-cause pair extraction?

A2: We investigate emotion-cause pair distribution and extraction performance for two consecutive utterances from the same speaker, under two conditions: when emotion changes and remains consistent. As illustrated in Figure 7, the experimental results reveal that the introduction of the label constraint strategy yields a significant improvement, of more than 2 percent, on instances where emotion changes. On the other hand, the improvement on instances where the emotion remains the same is relatively moderate. These findings conclusively confirm that our label constraint mechanism effectively captures emotion transition cues and significantly enhances extraction performance by paying more attention to the middle utterance as a cause.

Q3: How does the parameter in adjusted softmax β influence the result?

A3: In the adjusted softmax function as defined in Eq. (25), the JS divergence of the sentiment vector is utilized to assess the adjustment degree required for the softmax function. Given that the JS divergence typically yields relatively small values, the enhancement for positive samples becomes constrained. Consequently, the incorporation of a β parameter is essential to achieve optimal performance. The impact of β on overall performance is crucial for the effectiveness of our label constraint module. As illustrated in Figure 8, the variation in F1 score across different β values is depicted, where $\beta = 0$ equates to applying the conventional softmax function—effectively removing the label constraint module. Initially, as β increases, there is a gradual improvement in the F1 score, underscoring the beneficial role of the β parameter. Nevertheless, performance reaches its peak at $\beta = 5$ and subsequently diminishes. Such a decline is attributable to an excessively high β value, which over-adjusts the softmax function, leading to an increased generation of positive examples and, in turn, impairing the model's judgment capabilities. It is noteworthy that the model's effectiveness significantly declines when β surpasses 10. In summary, after thorough evaluation of how different β values affect our model's performance, we have determined that setting the hyperparameter β to 5 optimizes outcomes.

Table 5. Experimental results comparing with and without RoPE embedding. The *Prop* column denotes the proportion of emotion-cause pairs with the corresponding distance .

Distance		P	R	F	Prop. (%)
W/o RoPE	all	48.74	60.06	53.81	100.00
	0	58.98	76.48	66.60	53.11
	1	44.45	60.11	51.11	29.63
	2	21.57	32.50	25.93	9.38
	>3	19.23	14.08	16.26	7.88
W/i RoPE	all	51.78	59.69	55.45 (**+1.64**)	100.00
	0	59.26	79.51	67.91 (**+1.31**)	53.11
	1	44.33	60.11	51.03 (-0.08)	29.63
	2	24.08	20.62	22.22 (-3.71)	9.38
	>3	25.00	4.10	7.05 (-9.21)	7.88

Q4: To what extent does RoPE influence the performance of emotion-cause pair extraction at varying distances?

A4: We investigate the impact of introducing RoPE on the performance of emotion-cause pair extraction across various pairwise distances. As depicted in Table 5, a gradual decrease in extraction performance is observed as the distance between utterances increases. This trend can be attributed to the likelihood that utterances at larger distances are less prone to forming an emotion-cause pair, a notion corroborated by the data shown in the last column of the table, where over half of the emotions are identified as self-caused.

Interestingly, the incorporation of RoPE significantly enhances the extraction of emotion-cause pairs within a single utterance by 1.31 percent. However, for pairs spanning longer distances, performance declines compared to models not utilizing RoPE, and this decrement becomes more pronounced at greater distances. This phenomenon suggests that RoPE is particularly effective at capturing distance-sensitive information, thereby enhancing performance predominantly for proximal utterance pairs, where the formation of emotion-cause pairs is more likely. Conversely, at extended distances, the effectiveness of RoPE diminishes. Nonetheless, an overall performance improvement of 1.61 percent is noted, primarily attributed to the high incidence of closely situated emotion-cause pairs.

5.9 Case Study

We delved into specific examples via case studies to illustrate the efficacy of our HiLo model, especially highlighting the importance of the holistic interaction and label constraint modules. Refer to Figure 9, which showcases two distinct scenarios: the prediction results of the ECPE under the full model and after the removal of the holistic interaction or label constraint.

Left Scenario: *Chandler* speaks loudly in utterance #8. Subsequently, *Rachel*, expressing *anger*, comments, “Who is being loud?” in utterance #10. This indicates that the heightened audio from utterance #8 serves as a causative factor for *Rachel*’s emotion in utterance #10. The recognition of this emotion-cause relationship hinges on the interaction between the audio of utterance #8 and the content of utterance #10. Impressively, our HiLo model correctly identifies the emotion-cause pair (#10, #8). However, a model without the holistic interaction module fails in this regard. This instance underscores the capability of our model to perform intricate cross-utterance and cross-modality interactions, thereby enhancing the accuracy of emotion-cause pair detection.

Right Scenario: In another scene, *Phoebe*’s *sadness* in utterance #16 stems from her inability to find a cat. This emotional tone shifts when *Rachel* announces in utterance #17 that she has located the cat, leading *Phoebe* to express *joy* in utterance #18. Given this progression, it is evident that utterance #17, serving as the solitary bridge between the two emotional states, acts as the catalyst for *Phoebe*’s transition from sadness to joy. Our HiLo model

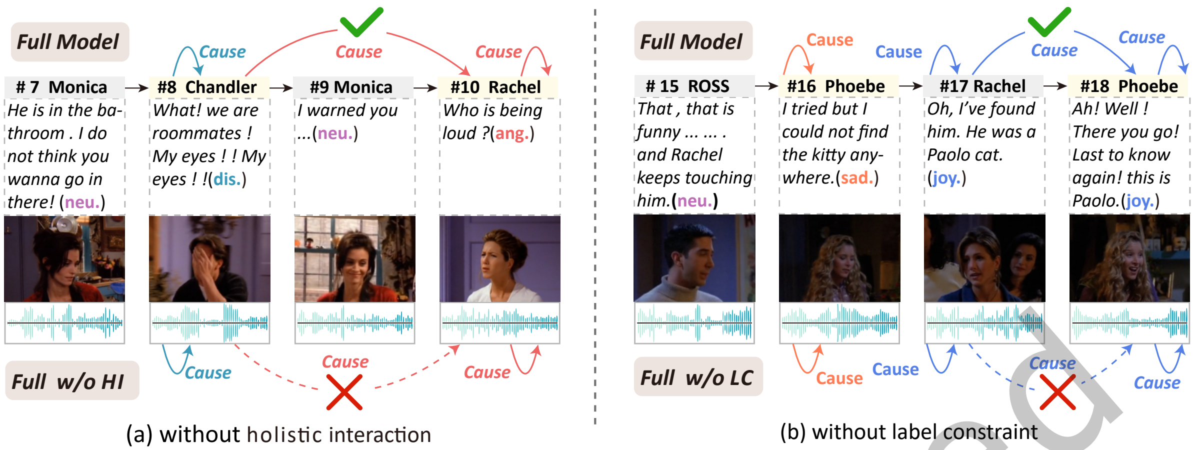


Fig. 9. Illustration of the impact of holistic interaction and label constraint in emotion-cause pair extraction. The absence of holistic interaction prevents the model from grasping intricate interactions, leading to missed modality-sensitive emotion-cause pairs. Meanwhile, without the label constraint, the model struggles to detect causes for utterances that show pronounced emotion transitions.

adeptly captures this shift in emotions, recognizing utterance #17 as the causative factor for *Phoebe*'s change from *sadness* in utterance #16 to *joy* in utterance #18. Conversely, when the label constraint is omitted, the model struggles to establish this connection. This illustrates the critical role of the label constraint in tracing emotional trajectories and identifying causative links.

6 CONCLUSION

In this paper, we proposed a model integrating holistic interaction and label constraint mechanisms for the MECPE task. We identified that existing MECPE models have not fully considered the role of feature interaction and emotional transition, leading to subpar performance. To remedy these shortcomings, we employed a holistic interaction via multi-head self-attention to enhance both utterance representation and relationships between utterances. In the emotion-cause extraction stage, we introduced a label constraint feature into the pair comparison process to capture emotion transition information. Additionally, we incorporated the RoPE relative distance embedding to enable distance-sensitive pair extraction. We evaluated our model using a standard benchmark, and the results showed that our approach outperforms the best baseline model by around 2 percent in F1 scores. Detailed analysis further confirmed the rationality and effectiveness of each proposed module. In future work, we plan to extend the holistic interaction and label constraint mechanisms to more tasks, aiming to explore their potential benefits in the realm of multimodal content analysis.

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